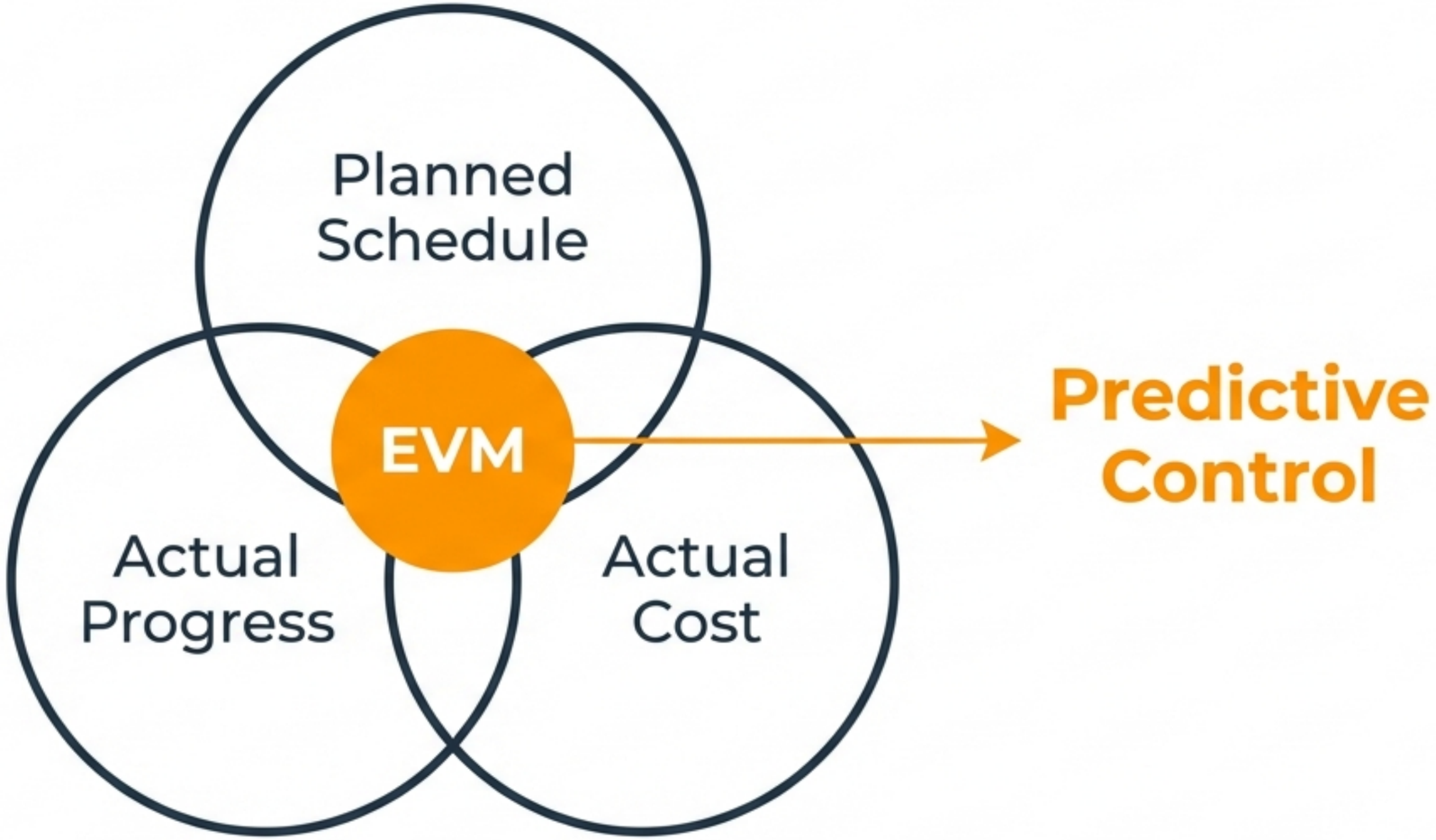
LECTURE 12 — Cost Management & Earned Value (EXAM FAVORITE)

- Cost + Schedule Integration
- Plan vs. Reality



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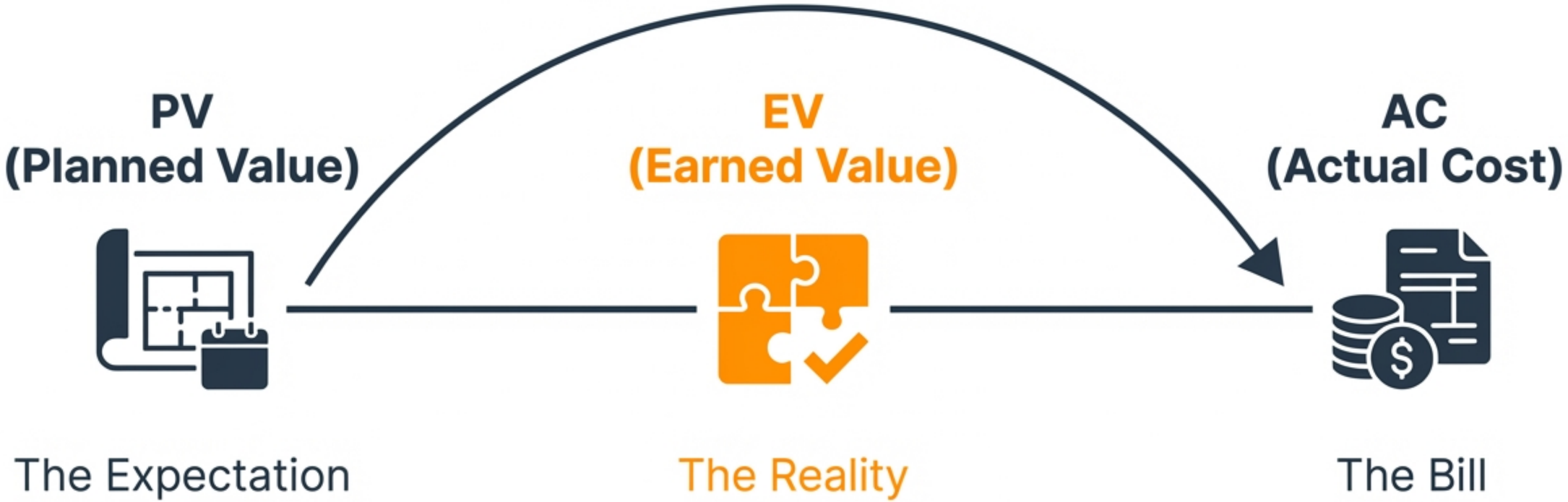
The 3 Core Metrics



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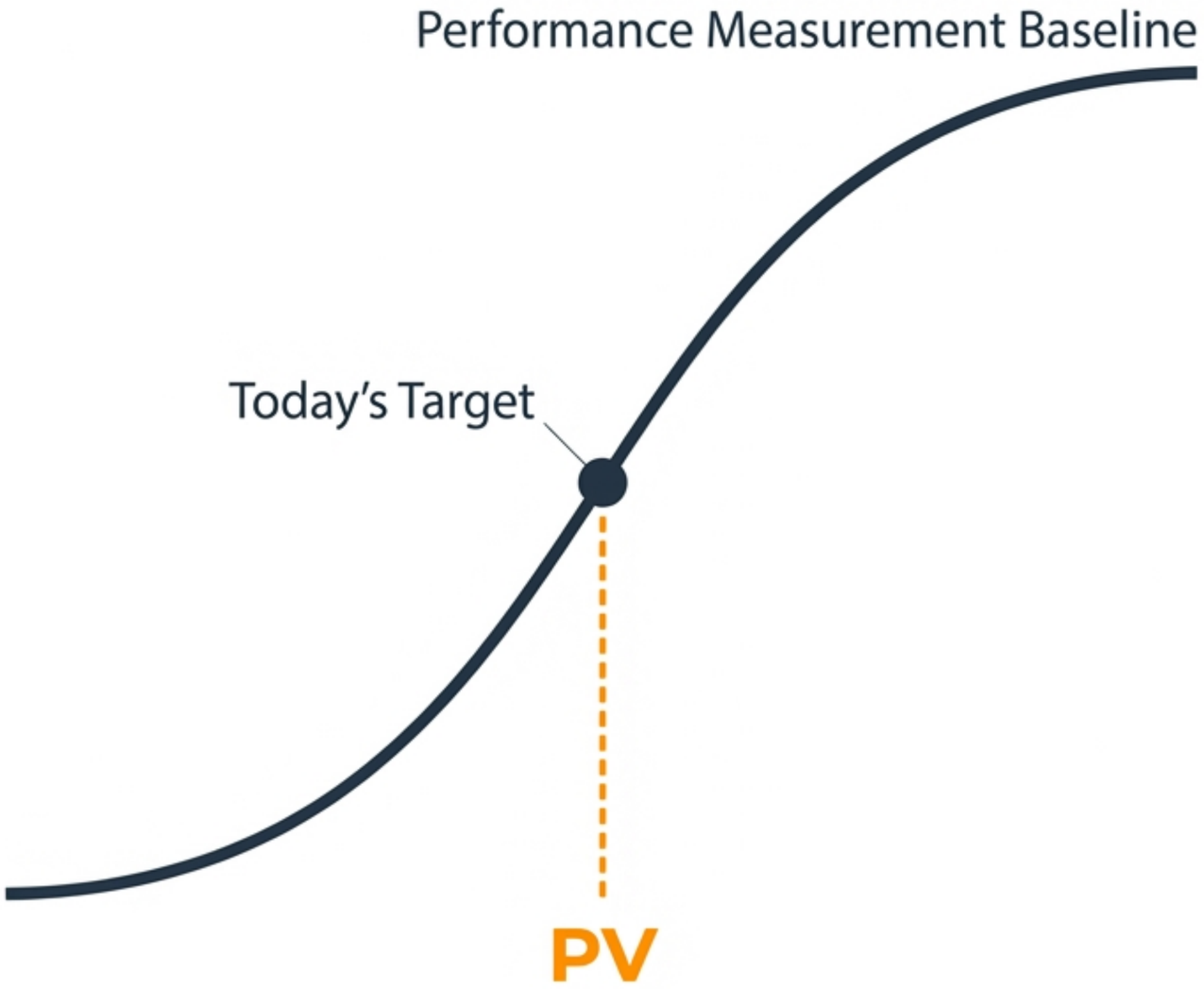
Planned Value (PV)

Budgeted Work Scheduled

Progress Expectation

Baseline Reference



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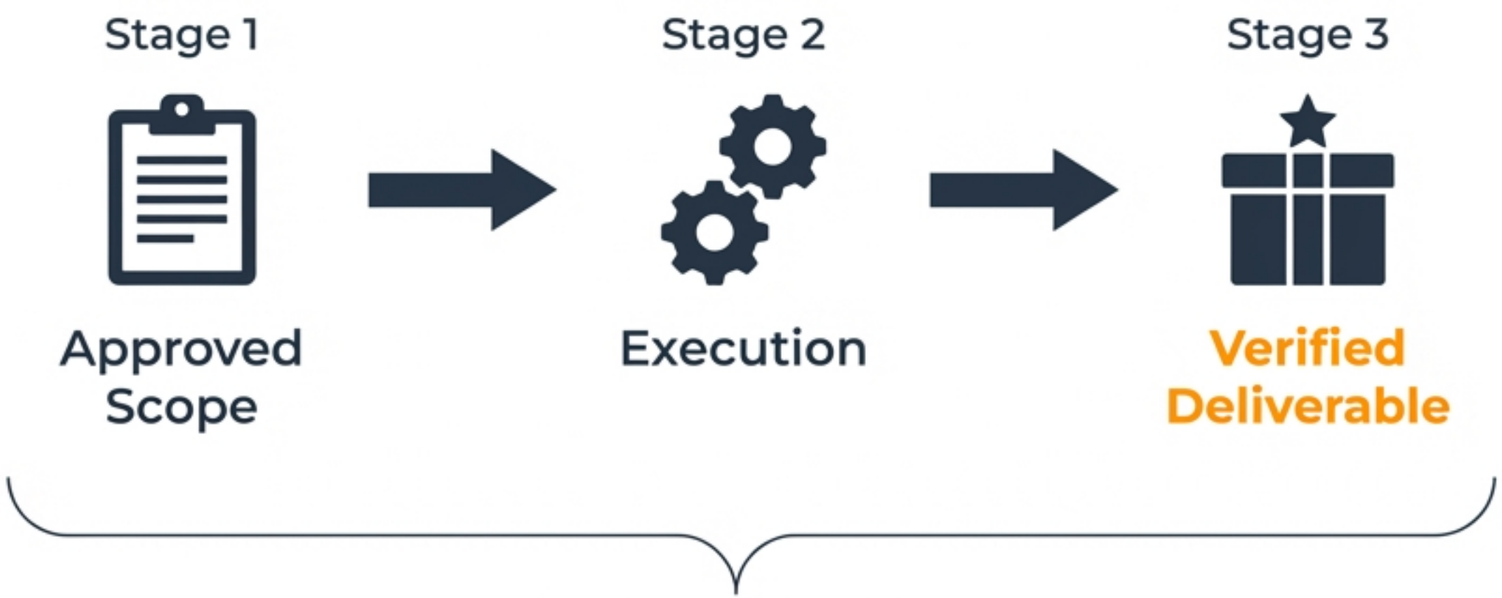
Earned Value (EV)

Work Completed

Value Realized

True Progress Measure



EV

Realized Value

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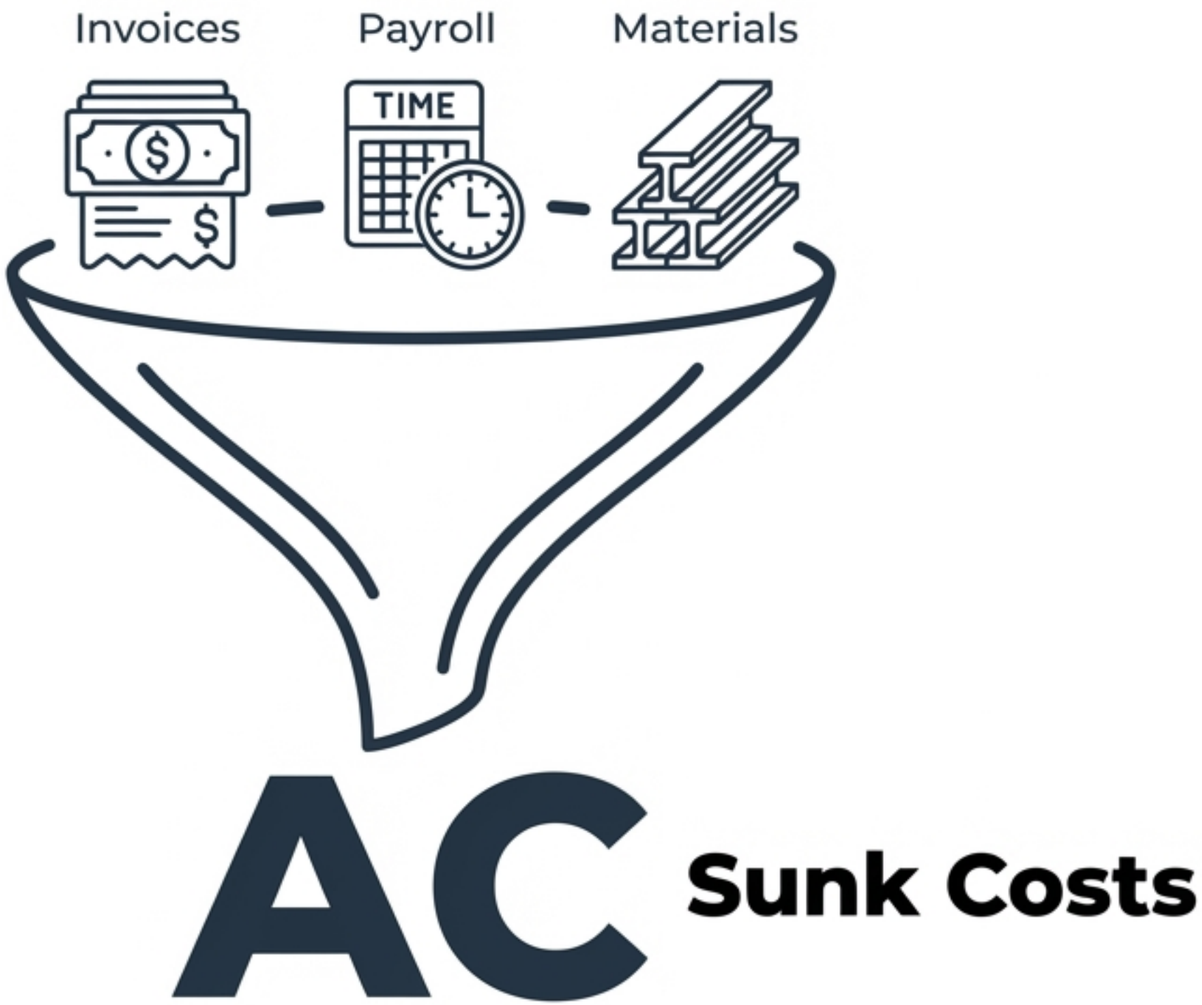
Actual Cost (AC)

Capital Expended

Real Cost Basis

Zero Estimation



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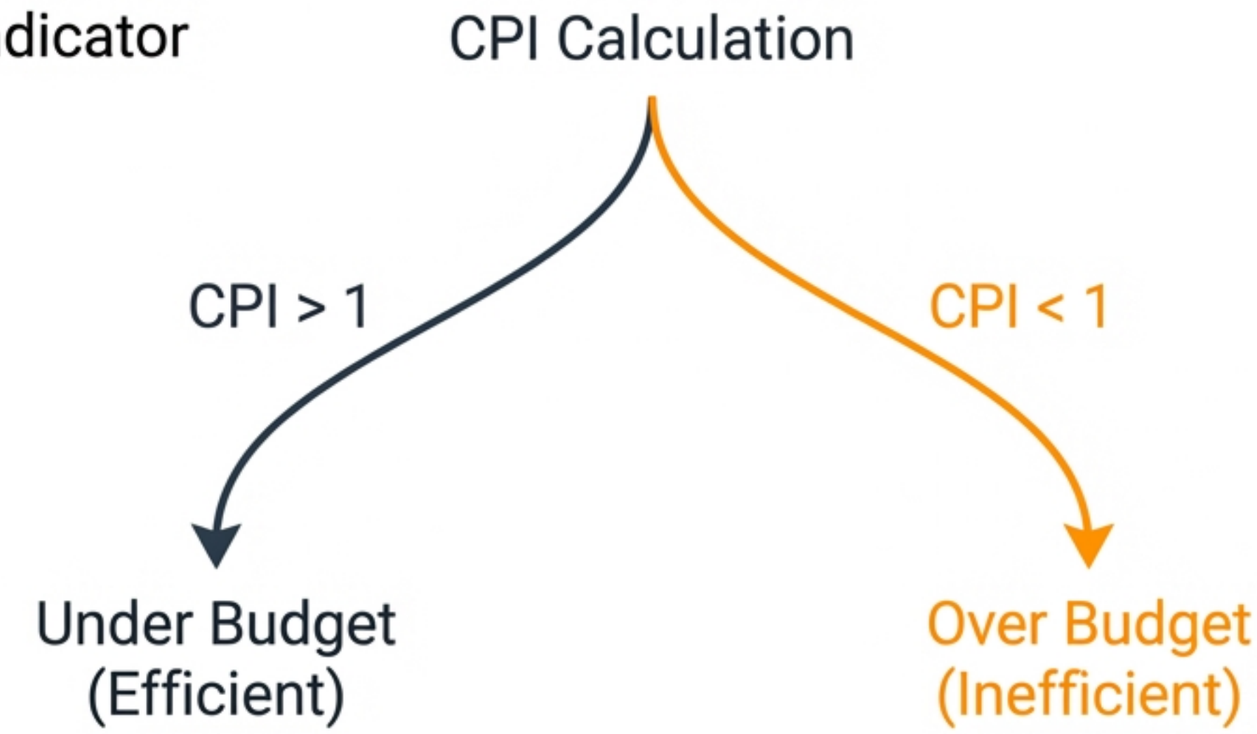
Cost Performance Index (CPI)

CPI = EV / AC

Efficiency Ratio

Financial Health Indicator



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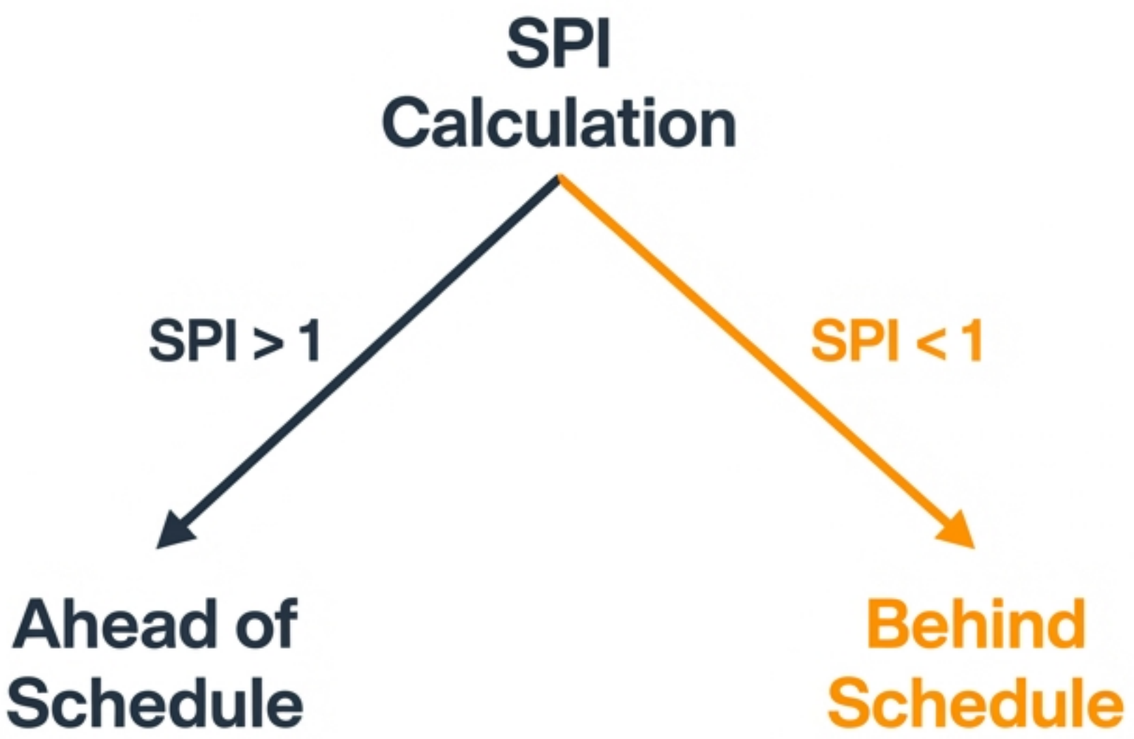
Schedule Performance Index (SPI)

SPI = EV / PV

Velocity Ratio

Timeline Health Indicator



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Forecasting (EAC)

Estimate at Completion
Future Cost Prediction
Driven by Current Trends

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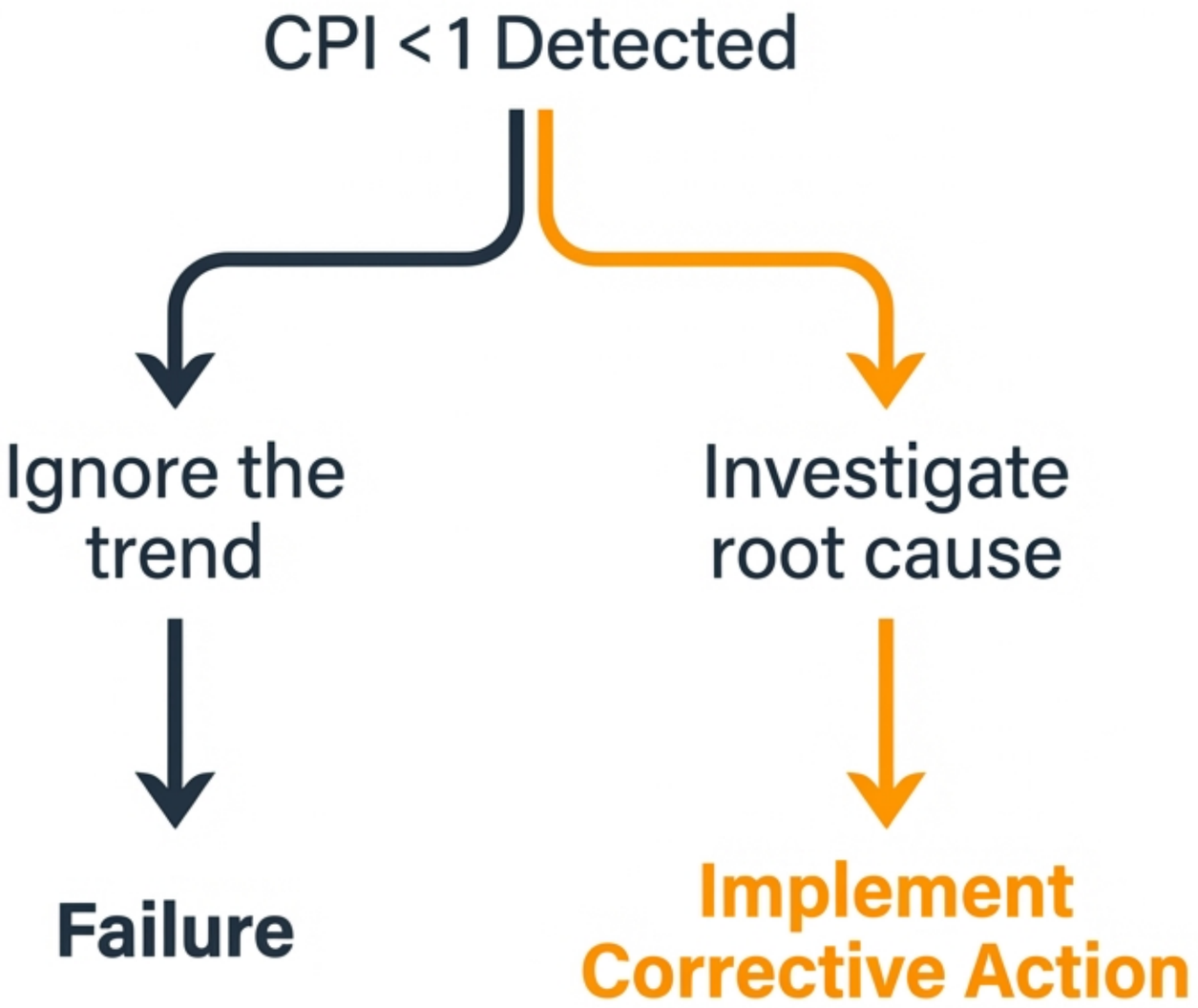
Action Scenario: Budget Overrun

- Situation: CPI < 1
- **Financial Trend:** Over Budget
- PM Action Required



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Overcoming Exam Traps

Distractor vs. Reality | Ratios Matter | Interpretation Over Calculation

Common Traps		PMI Mindset
Confusing EV & PV	→	**EV is delivered value**
Ignoring CPI/SPI ratios	→	**Ratios dictate action**
Memorizing blind formulas	→	**Interpret the trend**

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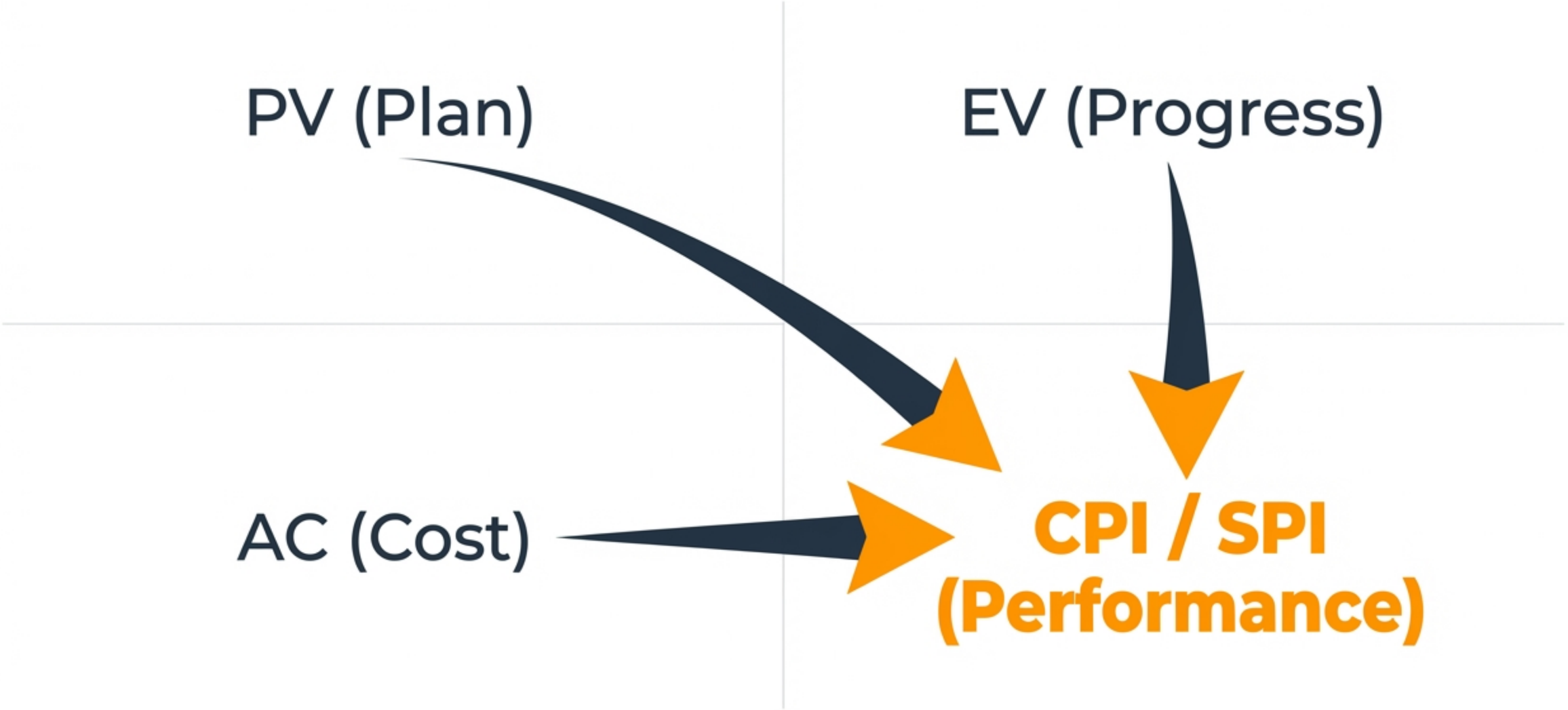
EVM Mastery Synthesis



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PMP Exam Scenario

Scenario Analysis

Project Variance

Decision Making

A project has a CPI of 0.8 and an SPI of 1.2.

↓

What is the project status?

A. Under budget and ahead of schedule.

B. Over budget and behind schedule.

C. Over budget and ahead of schedule.

D. Under budget and behind schedule.

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